

# MOSTOFA HASIN MAHDI

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## EDUCATION

**Shanto-Mariam University of Creative Technology**  
*Bachelor of Science in Computer Science and Engineering*  
CGPA: 3.96

Dhaka, Bangladesh  
**2022 - Expected Nov 2026**

## PROJECTS

**NAB Preg AI** *FastAPI, Next.js, XGBoost, Scikit-Learn, MultiOutputClassifier, Mistral AI, Supabase*  
Lead AI Engineer (Team of 5)

- Built a **single-pass ML model** predicting 3 clinical risk targets in **under 50ms**, using WHO-aligned thresholds to correct class imbalance.
- Developed an **OCR pipeline** integrating Mistral's **Pixtral-12B vision model** via a FastAPI proxy, parsing clinical prescriptions and lab reports into sanitized JSON payloads for frontend validation.
- Engineered a **5-step OCR failover** pipeline cycling through 4 backup Mistral API keys before routing to a local client-side Tesseract.js parser as a final fallback.
- Implemented dual-layer validation using Zod (client) and Pydantic (server) to sanitize 10 distinct patient fields, preventing malformed payload insertion into the database.
- *Demo:* [Live Link](#) | [GitHub Link](#)

**PDF to Markdown Conversion Engine** *PyMuPDF, pdfplumber, Pandas, Tabulate, spaCy, PyTesseract*  
Solo Developer

- Developed an open-source Python CLI tool leveraging a custom layout-detection algorithm with coordinate bounding boxes to accurately sort text, detect headings, and prevent overlapping content.
- Integrated **NLP-based text cleaning** (spaCy) to resolve sentence fragmentation, and implemented an **OCR fallback system** (PyTesseract) to process scanned pages as high-resolution images.
- Built an algorithmic table extraction tool that identifies complex grids and reconstructs them into aligned Markdown tables.
- *Demo:* [GitHub Link](#) | [Release Link](#)

**Bizit Business Suite** *React, FastAPI, PostgreSQL, Neon DB, Render*  
Solo Developer

- Isolated data across **5 test tenants** in a single PostgreSQL schema using Row-Level Security (RLS) to support concurrent access with **zero cross-tenant leaks**.
- Built a **3-tier RBAC system** (Admin/Manager/Viewer) and utilized Pydantic validation across 24 API endpoints to ensure malformed payloads never reach the database.
- Designed real-time finance dashboards that aggregate sales, costs, and profit in **under 500ms**, replacing the manual spreadsheet workflow which takes business owners 30 minutes per day.
- *Demo:* username: hasin, password: hasin123 | [Live Link](#) | [GitHub Link](#)

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, C++  
**Frameworks:** React, Next.js, FastAPI  
**AI/ML:** XGBoost, Pandas, Scikit-Learn, OCR (Mistral Pixtral, PyTesseract), NLP (spaCy)  
**Databases:** PostgreSQL (Supabase, Neon), MongoDB, MySQL  
**Testing/DevOps:** Postman, Vercel, Render, SwaggerUI, GitHub Actions  
**Tools:** Git, GitHub, Docker, Linux Shell Scripting

## ACHIEVEMENTS

- **Hackathons & Showcases:** Finalist at CloudcampBD Infinity AI BuildFest 2026, Champion of SMUCT Software Project Showcase 2026.
- **Programming Contests:** Second Runner-up at SMUCT CSE FEST V2.0 Programming Contest, Participated in BUET & SUST IUPC Prelims, ICPC Prelim Regional Contests (2022–2023).